

Abhraneel Sarma

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Employment

2025- **Postdoctoral Researcher**
Institute of Human-Centered Computing, Graz University of Technology
Advisor: Alex Lex

Education

2019-2025 **PhD, Computer Science**
Northwestern University
Advisors: Matthew Kay, Jessica Hullman

2019-2021 **Master of Science, Computer Science**
Northwestern University

2016-2018 **Master of Science, Information**
University of Michigan
Advisor: Matthew Kay

2012-2016 **Bachelors, Design**
Indian Institute of Technology Guwahati

Awards and Honors

Papers

2024 **Best Paper Honorable Mention** for “Odds and Insights ...”, ACM CHI
2024 **Best Paper Honorable Mention** for “Opportunities, tensions ...”, ACM DIS
2023 **Best Paper Honorable Mention** for “multiverse ...”, ACM CHI
2022 **Best Paper Honorable Mention** for “Evaluating the use of ...”, IEEE VIS
2019 **Best Paper Award** for “Increasing the Transparency of ...”, ACM CHI

Other Activities

2021-2025 **Special Recognitions for Outstanding Reviews** (five), ACM CHI

Publications

Peer-reviewed Full-length Conference and Journal Articles

C12 2026 **Should We Dangle a Carrot? The Effect of Performance-based Incentives in Visualization Experiments**
[Abhraneel Sarma](#) , Matthew Kay, Sheng Long, Michael Correll and Alexander Lex
IEEE Transactions on Visualization and Computer Graphics (proc. VIS 2026)

- C11 2026 **Adapting CCDF Plots for Visualizing Ordinal Regression Results**
[Abhraneel Sarma](#)
 IEEE Transactions on Visualization and Computer Graphics (proc. VIS 2026)
- C10 2025 **More Forecasts, More (Decision) Problems: How Uncertainty Representations for Multiple Forecasts Impact Decision Making** [DOI](#)
[Abhraneel Sarma](#) , Maryam Hedayati and Matthew Kay
 Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI)
- C9 2024 **Odds and Insights: Decision Quality in Exploratory Data Analysis Under Uncertainty** [DOI](#)
[Abhraneel Sarma](#) , Xiaoying Pu, Yuan Cui, Eli T Brown, Michael Correll and Matthew Kay
 Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI)
Best Paper Honorable Mention
- C8 2024 **Milliways: Taming Multiverses through Principled Evaluation of Data Analysis Paths** [DOI](#)
[Abhraneel Sarma](#) , Kyle Hwang, Jessica Hullman and Matthew Kay
 Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI)
- C7 2024 **Opportunities, tensions, and challenges in computational approaches to addressing on-line harassment** [DOI](#)
 Evey Huang, [Abhraneel Sarma](#) , Sohyeon Hwang, Eshwar Chandrasekharan and Stevie Chancellor
 Proceedings of the 2024 ACM Designing Interactive Systems Conference (DIS)
Best Paper Honorable Mention
- C6 2024 **The multiverse of universes: A tutorial to plan, execute, and interpret multiverses analyses using the R package multiverse** [DOI](#)
 Martin Götz, [Abhraneel Sarma](#) and Ernest O’Boyle
 International Journal of Psychology
- C5 2023 **multiverse: Multiplexing Alternative Data Analyses in R Notebooks** [DOI](#)
[Abhraneel Sarma](#) , Alex Kale, Michael Moon, Nathan Taback, Fanny Chevalier, Jessica Hullman and Matthew Kay
 Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI)
Best Paper Honorable Mention
- C4 2022 **Evaluating the Use of Uncertainty Visualisations for Imputations of Data Missing At Random in Scatterplots** [DOI](#)
[Abhraneel Sarma](#) , Shunan Guo, Jane Hoffswell, Ryan Rossi, Fan Du, Eunye Koh and Matthew Kay
 IEEE Transactions on Visualization and Computer Graphics (proc. VIS 2022)
Best Paper Honorable Mention
- C3 2021 **An automated approach to reasoning about task-oriented insights in responsive visualization** [DOI](#)
 Hyeok Kim, Ryan Rossi, [Abhraneel Sarma](#) , Dominick Moritz and Jessica Hullman
 IEEE Transactions on Visualization and Computer Graphics (proc. VIS 2021)
- C2 2020 **Prior Setting in Practice: Strategies and Rationales Used in Choosing Prior Distributions for Bayesian Analysis** [DOI](#)
[Abhraneel Sarma](#) and Matthew Kay
 Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI)

- C1 2019 **Increasing the Transparency of Research Papers with Explorable Multiverse Analyses** DOI
 Pierre Dragicevic, Yvonne Jansen, Abhraneel Sarma , Matthew Kay and Fanny Chevalier
 Proceedings of the ACM CHI Conference on Human Factors in Computing Systems (CHI)
Best Paper Award

Workshops and Lightly-Reviewed Articles

- W2 2024 **Tasks and Telephones: Threats to Experimental Validity due to Misunderstandings of Visualisation Tasks and Strategies**
Abhraneel Sarma , Sheng Long, Michael Correll and Matthew Kay
 evaluation and BEyond - methodoLogIcal approaches for Visualization (BELIV) Workshop at IEEE VIS
- W1 2019 **Interactive Visualizations Tools for Prior Setting In Bayesian Analysis: Challenges For Evaluation**
Abhraneel Sarma and Matthew Kay
 Workshop on Human-Centered Study of Data Science Work Practices at ACM CHI

Teaching

Instructor of Record

- 2025 **CS333 Interactive Information Visualization**
 Northwestern University

Conference Courses

- 2022, 2023 **Transparent Practices for Quantitative Empirical Research**
 Co-instructors: Chat Wacharamanotham, Fumeng Yang, Xiaoying Pu, Lace Padilla, Maryam Hedayati (this is a lecture series with rolling instructors).
 ACM CHI 2022 (online)
 ACM CHI 2023 (hybrid)
 IEEE VIS 2023 (in-person)

Teaching Assistantship

- 2020 **JOUR377 Data Analysis and Visualization**
 Northwestern University
- 2018 **SI330 Data Manipulation in Python**
 University of Michigan
- 2017 **SI588 Fundamentals of Human Behavior**
 University of Michigan

Mentoring

Northwestern University

- 2024-2026 **Yong-yu Huang**, undergraduate student for “Improving Trust by Communicating Uncertainty Due to Non-Response in Election Polls”
- 2024 **Sheng Long**, PhD student, co-mentored with Matthew Kay for “Tasks and Telephones...”
- 2024 **Mandi Cai**, PhD student, informal research mentoring
- 2021-2023 **Kyle Hwang**, undergraduate student for “Milliways ...”
- 2022 **Philip Clement**, undergraduate student for “Milliways ...”

Service

Program Committee

- 2026 **Program Committee Member, IEEE VIS**
- 2026 **Program Committee Member (Short Papers), EuroVis**
- 2024- **Journal of Visualization and Interaction, Open Practices Chair**

Paper Reviews

- 2024-2025 **The ACM Symposium on User Interface Software and Technology (UIST)**
- 2024-2025 **Information Visualization**
- 2023-2025 **IEEE Transactions on Visualization and Computer Graphics (TVCG)**
- 2022 **Workshop on TRust and EXpertise in Visualization (TREX VIS)**
- 2021-2023 **IEEE Visualization and Visual Analytics Conference (VIS)**
- 2021-2025 **The ACM Conference on Human Factors in Computing Systems (CHI)**

At Northwestern University

- 2022 **Computer Science Incoming PhD Student Orientation, Student Coordinator**
- 2020-2023 **Computer Science Social Initiative, Organising Committee**

Student Volunteering

- 2017-18 **IEEE Visualization and Visual Analytics Conference (VIS)**

Talks & Panels

Invited Talks

- 2025 **Designing Interfaces to Support Reliable Data-driven Decision-making**
Department of Computer Science, Washington University in St. Louis
- 2021 **Using the multiverse R package for Sensitivity Analysis**
Department of Engineering Sciences and Applied Mathematics, Northwestern University
- 2019 **Statistics and Scientific reporting: Can we do better?**
ICOS Big Data Summer Camp, University of Michigan

Discussion Panels

- 2024 **"Navigating Graduate School" Panel**
with Federico Sossai, Radhika Garg, Shubham Shahi and Ayse Hunt
CS401: Introduction to Graduate Studies, Northwestern University
- 2024 **PhD Panel**
with Dietrich Geisler, Melissa Chen and Achala Mishra
CS298: Computer Science Research Track Program, Northwestern University
- 2024 **Student Research & FAQ Panel**
with Maryam Hedayati, Tommy McMichen, Donna Hooshmand, Aidan Fitzsimons, Maryam Azmandian and Nathaniel Hejduk
Computer Science Incoming PhD Student Orientation, Northwestern University

Research Assistantships & Internships

- 2019-2025 **Graduate Research Assistant**
MUCollective, Northwestern University
- 2023 **Research Intern**
Microsoft Corp.
Mentors: Jake Hofman, Dan Goldstein
- 2021 **Research Intern**
Adobe, Inc
Mentor: Shunan Guo
- 2016-2019 **Graduate Research Assistant (full time, May 2018 onwards)**
MUCollective, University of Michigan
- 2017 **User Experience Intern**
Office of Academic Innovation, University of Michigan
- 2015 **Undergraduate Research Intern**
Keio-NUS CUTE Center, National University of Singapore